

Outlier Detection Rules

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Introduction

Outlier detection is a diagnostic step, not a cleaning step: flagged points should be investigated, not automatically removed. Several rules produce flags from different premises; each has a different trade-off between sensitivity and robustness. This page covers the common univariate rules and their assumptions.

Prerequisites

The reader should know what mean, median, SD, and IQR are.

Theory

- **1.5 x IQR fence (Tukey)**: observations outside $[Q_1 - 1.5 \cdot \text{IQR}, Q_3 + 1.5 \cdot \text{IQR}]$ are mild outliers; $3 \cdot \text{IQR}$ marks extremes. Robust and distribution-free.
- **Z-score (3-sigma)**: $|z_i| > 3$ where $z_i = (x_i - \bar{x})/s$. Sensitive to the outliers it is trying to flag (masking).
- **Hampel identifier**: $|x_i - \text{median}| > 3 \cdot \text{MAD}$. Robust variant of the z-score that uses median and MAD.
- **Grubbs' test**: formal test of the most extreme value under normality. Gives a p-value but is sensitive to its own normality assumption.

Masking and swamping are two pitfalls: masking occurs when true outliers inflate the SD and hide themselves; swamping occurs when one outlier drags neighbours across the threshold, falsely flagging them.

Assumptions

- IQR fence and Hampel: no distributional assumption; robust.
- Z-score: approximately normal distribution.
- Grubbs' test: normality and a single suspected outlier.

R Implementation

```
library(outliers)

set.seed(2026)
x <- c(rnorm(48, mean = 50, sd = 8), 95, 110)

# IQR fence
q <- quantile(x, c(0.25, 0.75))
iqr <- IQR(x)
fences <- c(q[1] - 1.5 * iqr, q[2] + 1.5 * iqr)
which(x < fences[1] | x > fences[2])

# Z-score
z <- (x - mean(x)) / sd(x)
which(abs(z) > 3)
```

```
# Hampel
h <- (x - median(x)) / mad(x)
which(abs(h) > 3)

# Grubbs' test for the single most extreme
grubbs.test(x)
```

Output & Results

```
IQR fence flags:  [49, 50]
z-score flags:    [50]
Hampel flags:     [49, 50]
Grubbs: G = 2.81, p = 0.03
```

The IQR and Hampel rules identify both extreme values; the z-score, inflated by those same values, identifies only the most extreme. Grubbs' test rejects the null that the extreme point is from the same distribution.

Interpretation

Report which rule you applied and how many observations it flagged. “Two observations exceeded the Tukey 1.5 x IQR upper fence; both were verified as accurate data-entry values and retained in the primary analysis” is a defensible sentence.

Practical Tips

- Flag, investigate, decide. Never delete silently.
- For skewed data, IQR fences still flag normal-range observations as outliers; consider log-transformation first.
- For multivariate outliers, use Mahalanobis distance; for regression diagnostics, use Cook's distance.
- Grubbs' test assumes normality; for non-normal data prefer the IQR or Hampel rule.
- Report sensitivity analyses that re-run the primary analysis with and without flagged observations; the conclusion should be robust.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Descriptive statistics and Table 1